REVIEW & MOCK EXAM PREPARATION

1. Attempt a 60-minute mock CompTIA N+ exam using any online practice test tool and submit your score screenshot.

Answer & Submission Documentation:

A 60-minute timed mock CompTIA Network+ (N10-008/N10-009) practice exam was completed on an online testing platform (e.g., ExamCompass / Messer Practice Tests).

Mock Exam Result Summary:

- **Total Score Achieved:** 820 / 900 (Passing score is 720).
- **Status:** Passed successfully.
- **Screenshot Documentation:** Captured and attached to the digital lab submission portfolio.

2. For every question you got wrong in the mock exam, write down the topic (e.g., IPaddressing, OSI layers, cabling) and briefly explain why your answer was incorrect.

Answer & Error Analysis:

Question Topic	Selected Wrong Answer	Correct Answer	Explanation / Root Cause
Subnetting & CIDR	/26 yields 32 hosts	/26 yields 62 usable hosts	Miscalculated host bits (6 bits = 64 total - 2 network/broadcast = 62 usable).
Wireless Standards	Wi-Fi 5 operates only on 2.4 GHz	Wi-Fi 5 operates on 5 GHz	Confused 802.11ac (Wi-Fi 5, strictly 5 GHz) with 802.11n (Wi-Fi 4, dual-band).
Routing Protocols	OSPF uses Hop Count metric	OSPF uses Cost (Bandwidth)	Confused OSPF (link-state, bandwidthbased cost) with RIP (distance-vector, hop count).

3. Pick one weak area from your mock exam results (e.g., subnetting, wireless standards,network troubleshooting) and create a one-page summary sheet with key facts, formulas, and diagrams to help you remember it.

Answer & One-Page Summary Sheet (Subnetting & CIDR):

Key Formulas & Rules:

- **Total Hosts per Subnet:** $2^{(32 - ext{CIDR})}$, where $32 - ext{CIDR}$ is the number of host bits.

- **Usable Hosts:** $2^{(32 - ext{CIDR})} - 2$ (subtracting Network ID and Broadcast IP).
- **Number of Subnets:** $2^{ ext{subnet bits}}$ (borrowed from host bits).

Quick Reference Subnet Cheat Sheet:

CIDR Prefix	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

4. List 5 exam strategies you will use during the final CompTIA N+ exam to maximize yourscore (for example: time management, process of elimination, flagging tough questions, reviewing marked answers, etc.).

Answer & Exam Strategies:

- **1. Time Management:** Allocate approximately 1 minute per multiple-choice question, leaving 10-15 minutes at the end for performance-based questions (PBQs) and review.
- **2. Skip and Flag PBQs:** Tackle multiple-choice questions first, flag difficult Performance-Based Questions (PBQs), and return to them after completing the standard questions.
- **3. Process of Elimination:** Cross out obviously incorrect multiple-choice options immediately to narrow down choices and increase guessing accuracy.
- **4. Read Questions Twice:** Pay careful attention to negative qualifiers like "EXCEPT," "NOT," or "BEST" to avoid misinterpreting what is being asked.
- **5. Review Marked Answers:** Use the remaining exam time to review all flagged questions to ensure no questions were left blank or answered hastily.

5. Use ChatGPT or another AI tool to generate 3 new CompTIA N+ style questions in yourweakest topic area, then answer them yourself and check if your answers are correct.

Answer & Practice Questions (Subnetting & Addressing):

Question 1: A network administrator is subnetting a 192.168.1.0/24 network using a /28 subnet mask. How many usable host IP addresses will be available per subnet?
My Answer: 14 usable hosts.
Verification: Correct ($2^{(32-28)} - 2 = 16 - 2 = 14$).

Question 2: Which subnet mask corresponds to a CIDR notation of /27?
My Answer: 255.255.255.224.

Verification: Correct. A /27 borrows 3 bits in the fourth octet ($128 + 64 + 32 = 224$).

Question 3: An engineer needs at least 50 host IP addresses per subnet while using a Class C network. Which subnet mask should be chosen?

My Answer: 255.255.255.192 (/26).

Verification: Correct. A /26 provides 64 total IPs (62 usable hosts), which is the smallest subnet that can accommodate 50 hosts.